MIL-C-2399B

6 June 1966
SUPERSEDING
MIL-C-2399A
28 January 1960

MILITARY SPECIFICATION

CEMENT, LIQUID, TENT PATCHING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a liquid, synthetic rubber base cement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.

CCC-C-428 - Cloth, Duck, Cotton: Fire, Water, Weather and Mildew Resistant.

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 8040

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D-903-49 - Method of Test for Peel or Stripping Strength of Adhesives

(Application for copies of ASTM Standards should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Manufacturing Chemists' Association's Manual L-1

"Guide to Precautionary Labelling of Hazardous Chemicals"

(Application for copies of Manual L-1 should be addressed to Manufacturing Chemists' Association, Inc., 1825 Connecticut Avenue, N.W., Washington 9, D.C.)

3. REQUIREMENTS

* 3.1 Composition.- The adhesive furnished under this specification shall be based on an oil resistant elastomer in a solvent blend of petroleum naptha , toluol, xylol or acetone, methyl ethyl ketone or methyl iso butyl ketone. The cement shall contain a minimum of 30 percent total solids by weight when tested as specified in 4.3.

3.2 Consistency.- The cement shall have a consistency that will permit its ready application with a small stiff bristle brush, knife or flat object at 10°F. when tested as specified in 4.3.

3.3 Flexibility.- The cement bonded specimens, before aging and after aging, as specified in 4.4.3.1, shall not crack or show crease marks in the area of the bond when tested as specified in 4.3.

3.4 Adhesion.- The cement shall possess shear and stripping strength characteristics specified in table I when tested as specified in 4.3.



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TABLE I.- Adhesion requirements

| Characteristics | Requirement |
|---|---|
| Shear strength (minimum) | 100 pounds |
| Stripping strength: | |
| After 3 hours at room temperature (73.4 ± 2°F and 50 ± 4% R.H.) | 5 pounds |
| After 72 hours at room temperature (73.4 ± 2°F and 50 ± 4% R.H.) | 14 pounds |
| After 100 hours at 158°F. | 18 pounds |
| After 48 hours in water at room temperature (73.4 ± 2°F. and 50 ± 4% R.H.) | 12 pounds |

3.5 Stability.- The cement, after being conditioned as specified in 4.4.6, shall meet the shear and stripping strength requirements stipulated in table I. It shall not gel or thicken to an extent that will effect the consistency, specified in 3.2, when tested as specified in 4.3.

3.6 Workmanship.- The liquid tent patching cement shall be homogeneous, free from dirt or foreign matter and shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.



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4.2 Inspection.- Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item.-

4.2.2.1 Visual examination.- The packaged cement shall be examined for defects in workmanship, closure and marking. The sample unit shall be one packaged container of cement. The lot size for purposes of sampling shall be expressed in units of one packaged container of cement.

| Examine | Defect |
|---|---|
| Cement | Not homogeneous solution. |
| | Any dirt or foreign matter. |
| Filled container | Not type specified. |
| | Damaged (any dent, which may result in leakage of cement). |
| | Any unsound seam. |
| | Closure loose or damaged. |
| | Any spillage of cement. |
| | Exterior nonmetallic coating, not as specified (see 5.1.1). |
| Labelling or marking | Missing, incomplete or illegible. |
| | Size of lettering not as specified. |

* 4.2.2.2 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging and packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.



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| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted, incorrect, illegible, of improper size, location, sequence or method of application. |
| Materials | Any component missing. |
| | Any component damaged, affecting serviceability. |
| Workmanship | Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. |
| | Bulged or distorted container. |
| Content | Number of intermediate packages is more or less than required. |

4.2.2.3 Inspection levels and acceptable quality levels (AQL's).- The inspection levels for the purpose of determining the sample size and the acceptable quality levels (AQL's) expressed in defects per hundred units, shall be as follows:

| Examination paragraph | Inspection level | AQL |
|---|---|---|
| 4.2.2.1 | I | 4.0 |
| 4.2.2.2 | S-2 | 2.5 |

4.3 Testing of the end item.- The end item cement shall be tested in accordance with table II for the characteristics shown therein. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 8-ounces of cement. The lot size for purposes of sampling shall be expressed in units of one filled container of cement each. The sample size shall be as shown below for the lot size designated. The lot shall be unacceptable if one or more units fail to meet any requirement specified. For those characteristics where test results are reported as pass or fail, there shall be no evidence of failure of any test specimen to meet the requirements as specified.

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE II.- Instructions for testing

| Characteristic | Requirement paragraph | Test method | No. of determinations per sample unit | Results reported as: Pass or fail | Results reported as: Numerically to nearest |
|---|---|---|---|---|---|
| Identification of synthetic rubber | 3.1 | 1/ | | - | |
| Total solids | 3.1 | 4.4.1 | 2 | - | 0.5 percent |
| Consistency | 3.2 | 4.4.2 | 1 | X | |
| Flexibility: | | | | | |
| Unaged | 3.3 | 4.4.3.2 | 2 | X | |
| Aged | 3.3 | 4.4.3.1 | 2 | X | |
| Shear strength | 3.4 | 4.4.4 | 3 | - | pound |
| Stripping strength: | | | | | |
| After 3 hours at 73.4 ± 2°F | 3.4 | 4.4.5.3 | 3 | - | pound |
| After 72 hours at 73.4 ± 2°F | 3.4 | 4.4.5.4 | 3 | - | pound |
| After 100 hours at 158°F | 3.4 | 4.4.5.5 | 3 | - | pound |
| After 48 hours in water at 73.4 ± 2°F | 3.4 | 4.4.5.6 | 3 | - | pound |
| Stability (after conditioning) | 3.5 | 4.4.6 | | | |
| Consistency | 3.5 | 4.4.2 | 1 | X | |
| Shear strength | 3.5 | 4.4.4 | 3 | - | pound |
| Stripping strength: | | | | | |
| After 3 hours at 73.4 ± 2°F | 3.5 | 4.4.5.3 | 3 | - | pound |



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TABLE II.- Instructions for testing (cont'd)

| Characteristic | Requirement paragraph | Test method | No. of determinations per sample unit | Results reported as: Pass or fail | Results reported as: Numerically to nearest |
|---|---|---|---|---|---|
| Consistency (cont'd) | | | | | |
| Stripping strength: (cont'd) | | | | | |
| After 72 hours at 73.4 ± 2°F | 3.5 | 4.4.5.4 | 3 | - | pound |
| After 100 hours at 158°F | 3.5 | 4.4.5.5 | 3 | - | pound |
| After 48 hours in water at 73.4 ± 2°F | 3.5 | 4.4.5.6 | 3 | - | pound |

1/ Materials may be accepted on the basis of a supplier's certificate of compliance to the contracting officer for specified requirements.

4.4 Tests.-

4.4.1 Total solids.- Approximately 5 grams of liquid cement from a freshly opened sample container shall be poured as rapidly as possible into a tared cup, approximately 3 inches in diameter and 3/4 inch deep, equipped with a fitted cover. The cover shall be placed on the cup as rapidly as possible after pouring the cement and weight of the sample determined. Cover shall then be removed and the cup and sample exposed to a temperature of 70° ± 2°C. in a well ventilated oven until sample reaches constant weight. Replace cover on cup and cool in a desiccator and weigh. Percentage of total solids shall be calculated as follows:

$$\text{Percent total solids} = \frac{\text{Final weight of sample}}{\text{Initial weight of sample}} \times 100$$



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4.4.2 Consistency.- Condition an unopened container of cement for 5 hours at 10°F. At this same temperature, the container shall be opened and the cement applied to a 4 inch by 4 inch piece of cotton duck specified in 4.4.4.1 by means of a stiff bristle brush, knife or similar flat object. Observe ease of application, presence of undissolved material or evidence of gelation which makes application difficult.

4.4.3 Flexibility.-

4.4.3.1 Aging of specimens.- Specimen strips prepared as specified in 4.4.4.2 shall be placed in an oven maintained at 158° ± 2°F. for 100 hours, after which they shall be cooled to 73.4 ± 2°F. and tested as specified in 4.4.3.2.

4.4.3.2 Procedure.- Specimens prepared as specified in 4.4.4.2, shall be folded upon themselves 180° in the area of the bond. A roller, 2 inches in diameter and 2 inches wide, under a 10-pound load, shall be passed back and forth three times over the folded area. Return specimens to their unfolded condition and observe for evidence of crease marks or cracking in the area of the bond.

4.4.4 Adhesion (shear strength).-

4.4.4.1 Material for test.- Test specimens shall be made from cloth conforming to type III, 12.29 ounce of CCC-C-419 and shall be finished in accordance with CCC-C-428, type I or II, class 1.

4.4.4.2 Preparation for test specimens.- Two pieces of the duck specified in 4.4.4.1, 3-1/2 inches wide and 5-1/2 inches long shall be evenly and smoothly coated across the width at one end with a band of cement slightly more than one inch wide. Two coats of cement shall be applied as follows, the first coat shall be a volume of cement sufficient to give a calculated wet film thickness of 0.010 ± 0.002 inch. The cement shall be worked well into the cloth and allowed to air dry for 15 minutes before application of the second coat. The volume of the second coat of cement shall be calculated to give a wet film thickness of 0.020 ± 0.002 inch. The cemented ends shall be allowed to air dry for 15 minutes and shall then be pressed together lapping each other for a distance of one inch and having the uncoated ends extending in opposite directions. A roller, 2 inches in diameter and 2 inches wide, under a 10-pound load, shall be passed back and forth three times over the cemented area. After standing 72 hours a 1/4 inch strip shall be trimmed from each long edge leaving a specimen 3 inches wide by 10 inches long. This specimen shall then be further cut into three strips 1 inch wide by 10 inches long.



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4.4.4.3 Procedure.- The three strips prepared as specified in 4.4.4.2 shall each be pulled in a tensile testing machine equipped with minimum 2-inch width jaws 3 inches apart and having a jaw speed of 12 ± 0.5 inches per minute. The results reported for each sample unit shall be the average of the results obtained from the individual specimens tested.

4.4.5 Stripping strength.-

4.4.5.1 Preparation of specimens.- A piece of duck as specified in 4.4.4.1, 3-1/2 inches wide and 12 inches long (with the length in the warp direction), shall be coated on one side with cement in accordance with 4.4.4.2, except that each end is left uncoated to a depth of 2 inches, thereby giving a coated area 3-1/2 inches wide and 8 inches long. The specimen shall then be folded in half across the length, the two halves pressed together, rolled (in accordance with 4.4.4.2) and allowed to set for the specified time. A 1/4 inch strip shall be trimmed from each 3-1/2 inch side after which the central portion shall be cut lengthwise into 3 strips each 1 inch wide and 6 inches long.

4.4.5.2 Procedure.- The stripping strength shall be determined in accordance with the procedure of ASTM Method D-903-49 and the results calculated in accordance with this method.

4.4.5.3 Stripping strength (after 3 hours at room temperature).- Specimens prepared as specified in 4.4.5.1 shall be conditioned for 3 hours at 73.4 ± 2°F and then tested as specified in 4.4.5.2.

4.4.5.4 Stripping strength (after 72 hours at room temperature).- Specimens prepared as specified in 4.4.5.1 shall be conditioned for 72 hours at 73.4 ± 2°F and then tested as specified in 4.4.5.2.

4.4.5.5 Stripping strength (after 100 hours at 158°F).- Specimens prepared as specified in 4.4.5.1 shall be conditioned as specified in 4.4.3.1 and then tested as specified in 4.4.5.2.

4.4.5.6 Stripping strength (after 48 hours in water at 73.4 ± 2°F).- Specimens prepared as specified in 4.4.5.1 shall be immersed in water at room temperature for 48 hours and immediately tested as specified in 4.4.5.2.

4.4.6 Stability.- An unopened container of cement shall be soldered tightly shut and conditioned in an oven for 14 days at a temperature of 158 ± 2°F (70 ± 1.1°C). At the expiration of this period, the cement shall be tested as specified in 4.4.2, 4.4.4, 4.4.5.3, 4.4.5.4, 4.4.5.5 and 4.4.5.6 for determining conformance with applicable requirements as specified in table I.



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Cement shall be packaged in round, both ends double-seamed on, 1-pint or 1-gallon cans (see 6.2), conforming to type V, class 2 of PPP-C-96. Exterior nonmetallic coating shall be in accordance with plan B (see 6.2.1).

5.1.2 Level C.- Cement shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity for immediate use or controlled humidity storage. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Cement, packaged as specified in 5.1, shall be closed and packed in accordance with the applicable requirements in the Appendix of PPP-C-96.

5.2.2 Level B.- Cement, packaged as specified in 5.1, shall be closed and packed in accordance with the applicable requirements in the Appendix of PPP-C-96.

* 5.2.3 Level C.- Cement, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking.- Labelling shall comply with the provisions of the Manufacturing Chemists' Association's Manual L-1. In addition, the label shall contain the Manufacturer's name or trade mark, date of manufacture and shall also state in bold print that the adhesive may be used only in ventilated areas which have been approved by a Medical Officer.

## 6. NOTES

6.1 Intended use.- The liquid cement covered by this specification is intended for use in patching tentage as specified in applicable maintenance manuals.



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* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(c) Size of cans containing cement required (see 5.1.1).

* 6.2.1 The method of packaging referred to in 5.1.1 is similar to method III of MIL-P-116.

6.3 Toxicity.- Materials used in the formulation of the cement are toxic. Extreme care should be exercised in its use, in accordance with the instructions on the precautionary label.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL

Navy - WP

Air Force - 69

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Preparing activity:

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